

Given:

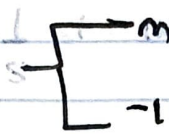
w_1



put $1/2$

put $1/2$

$c(m)$



$\text{if } m \geq 0$

$\text{if } m < 0$

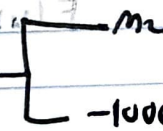
$$g_t(m_t, u_t) = c(m_t) + 2u_t, \quad g_N(m_N) = c(m_N)$$

let's start DP with terminal cost

at $t = N = 2$

~~$J_2^*(m_2)$~~

$G_2^*(m_2)$



$m_2 \geq 0$

$m_2 < 0$

$$J_2^*(m_2) = G_2^*(m_2)$$

at $t = N = 1$

using DP

$$J_1^*(m_1) = \min_{u_1} [c(m_1) + 2u_1 + J_2^*(m_2)]$$

$$= c(m_1) + 2u_1 + \min_{u_1} [J_2^*(m_1 + u_1 - w_1)]$$

~~$$\text{let } F(u_1) = \frac{2u_1 + 1}{2}$$~~

$$= c(m_1) + 2u_1 + \frac{J_2^*(m_1 + u_1 - 1)}{2} + \frac{J_2^*(m_1 + u_1 - 3)}{2}$$

$$\text{let } F(u_1) = 2u_1 + \frac{1}{2}(-1000(m_1 + u_1 - 1)) + \frac{1}{2}(-1000(m_1 + u_1 - 3))$$

$m_1 + u_1 < 1$

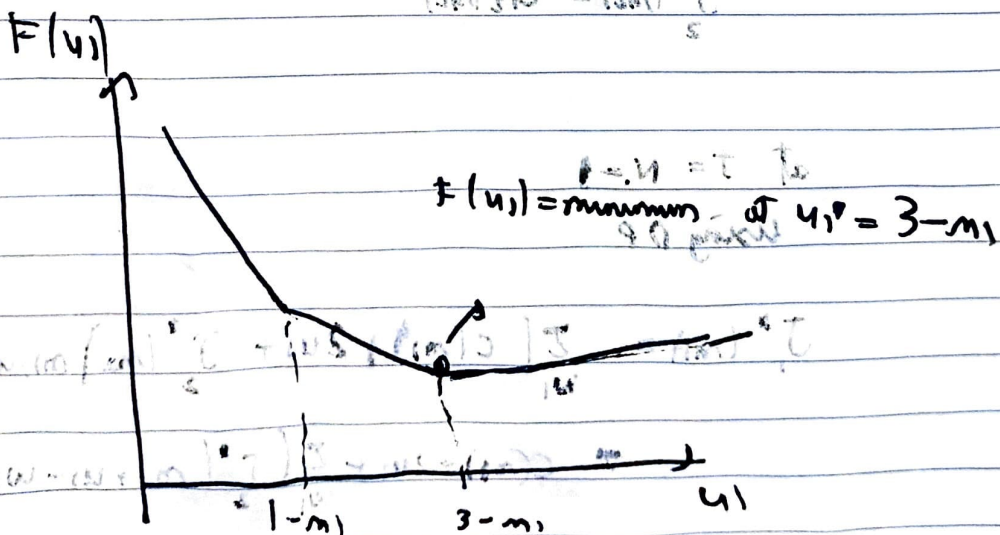
$$0 \leq m_1 \leq 1 \implies F(u_1) = 2u_1 + \frac{1}{2}(m_1 + u_1 - 1) + \frac{1}{2}(-1000(m_1 + u_1 - 3))$$

$$1 \leq m_1 + u_1 < 3$$

$$F(u_1) = 2u_1 + \frac{1}{2}(m_1 + u_1 - 1) + \frac{1}{2}(m_1 + u_1 - 3)$$

$$\text{for } m_1 + u_1 \geq 3$$

$$F(u_1) = \begin{cases} -998u_1 - 1000m_1 + 2000 & u_1 < 1 - m_1 \\ -995u_1 - 999m_1 + \frac{2999}{2} & 1 - m_1 \leq u_1 < 3 - m_1 \\ 3u_1 + m_1 - 2 & u_1 \geq 3 - m_1 \end{cases}$$



$$J'_1(m_1) = c(m_1) + 1 - 2m_1 \text{ at } u_1 = 3 - m_1$$

now

to solve sub problem

$$J_0(m_0) = \sum_{u_0} (c(m_0) + 2u_0 + J'_1(m_0, u_0))$$

$$= c(m_0) + 2u_0 + \sum_{u_0} (J'_1(m_0 + u_0 - u_0))$$

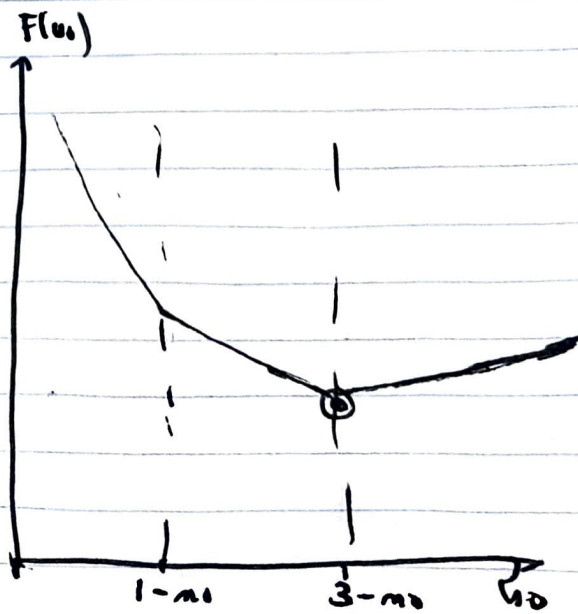
$$= c(m_0) + 2u_0 + \frac{1}{2} J_0''(m_0 + u_0 - 1) + \frac{1}{2} J_0''(m_0 + u_0 + 1)$$

$$J'_1(m_1) = \begin{cases} -1000m_1 + 7 - 2m_1 & m_1 < 0 \\ 7 - m_1 & m_1 \geq 0 \end{cases}$$

$F(u_0)$

$$F(u_0) = \begin{cases} m_0 + u_0 < 1 & 2u_0 + \frac{-1002(m_0 + u_0 - 1) + 7}{2} - \frac{1002(m_0 + u_0 - 3) + 7}{2} \\ 1 \leq m_0 + u_0 < 3 & 2u_0 + \frac{7 - (m_0 + u_0 - 1)}{2} - \frac{1002(m_0 + u_0 - 3) + 7}{2} \\ m_0 + u_0 \geq 3 & 2u_0 + \frac{7 - (m_0 + u_0 - 1)}{2} + \frac{7 - (m_0 + u_0 - 3)}{2} \end{cases}$$

$$F(u_0) = \begin{cases} -1000u_0 - 1002m_0 + 2011 & u_0 < 1 - m_0 \\ \frac{-999u_0 - 1003m_0 + 3021}{2} & 1 - m_0 \leq u_0 < 3 - m_0 \\ u_0 - m_0 + 9 & u_0 \geq 3 - m_0 \end{cases}$$



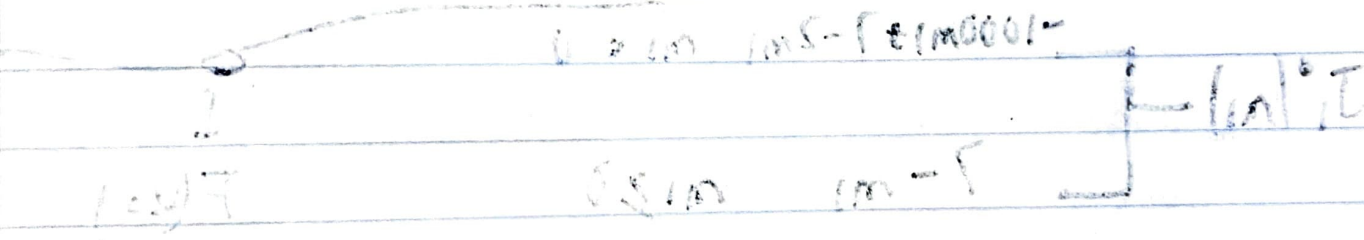
Optimum value occurs at

when

$$\left(\frac{d}{dm} [c(m) + 12 - 2m] \right) = 0 \Rightarrow c'(m) = 2$$

$$F'(m) = 12 - 2m = 0 \Rightarrow m = 6$$

$$F(6) = c(6) + 12 - 2(6) = 12$$



$$F(m) = c(m) + 12 - 2m$$

$$F'(m) = c'(m) + 12 - 2 = 0 \Rightarrow c'(m) = -10$$

$$c'(m) = -10 \Rightarrow c(m) = -10m + K$$

$$c(0) = 12$$

$$c(0) = 12 \Rightarrow K = 12$$

$$c(m) = -10m + 12$$

$$F(m) = -10m + 12 + 12 - 2m = -12m + 24$$